WORKED SOLUTION

STEP II 2010, Question 4

Pure • Integration by Substitution

STEP II 2010, Question 4

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/10-s2-q4

The key idea

The substitution $u = a - x$ swaps the ends of the interval and turns $f(x)$ into $f(a-x)$ and back again. So the integral equals its "partner", in which the numerator is the other half of the denominator. Adding the two versions makes the numerator equal the denominator, and the integrand becomes 1. The rest of the question is about spotting this structure in disguise. Part (ii) uses the same trick with the substitution $x \mapsto 1/x$, which swaps the ends of $[\frac{1}{2}, 2]$.

Solution

Part (i)

We assume throughout that $f(x) + f(a-x) \neq 0$ on $[0, a]$, so the integrand is defined. In I, substitute $u = a - x$, so $\mathrm{d}x = -\mathrm{d}u$. When $x = 0$, $u = a$, and when $x = a$, $u = 0$. Hence

$$I = \int_a^0 \frac{f(a-u)}{f(a-u)+f(u)}\,(-\mathrm{d}u) = \int_0^a \frac{f(a-u)}{f(u)+f(a-u)}\,\mathrm{d}u.$$

Since u is a dummy variable, we can rename it x, which gives the required second form of I.

Now add the two expressions for I:

$$2I = \int_0^a \frac{f(x)+f(a-x)}{f(x)+f(a-x)}\,\mathrm{d}x = \int_0^a 1\,\mathrm{d}x = a, \qquad \text{so} \qquad I = \frac{a}{2}.$$

The first integral.

? **Ask yourself** Where is the $f(x) + f(a-x)$ hiding in $\ln(2 + x - x^2)$?

The quadratic factorises: $2 + x - x^2 = (1 + x)(2 - x)$. So

$$\ln(2 + x - x^2) = \ln(1 + x) + \ln(2 - x),$$

which is valid because both factors are positive on $[0, 1]$. Take $a = 1$ and $f(x) = \ln(1 + x)$. Then $f(1 - x) = \ln(2 - x)$, and the integrand is exactly $\dfrac{f(x)}{f(x) + f(1 - x)}$. The denominator is never zero, since $2 + x - x^2 \geq 2$ on $[0, 1]$. Therefore

$$\int_0^1 \frac{\ln(x + 1)}{\ln(2 + x - x^2)}\,\mathrm{d}x = \frac{1}{2}.$$

The second integral. Expand the denominator with the compound-angle formula:

$$\sin(x + \tfrac{\pi}{4}) = \sin x \cos\tfrac{\pi}{4} + \cos x \sin\tfrac{\pi}{4} = \frac{\sin x + \cos x}{\sqrt{2}}.$$

So the integrand is $\sqrt{2}\,\dfrac{\sin x}{\sin x + \cos x}$. With $a = \frac{\pi}{2}$ and $f = \sin$, we have $f(\frac{\pi}{2} - x) = \cos x$. The denominator $\sin x + \cos x$ is positive on $[0, \frac{\pi}{2}]$. By the result above,

$$\int_0^{\pi/2} \frac{\sin x}{\sin(x + \frac{\pi}{4})}\,\mathrm{d}x = \sqrt{2} \cdot \frac{\pi/2}{2} = \frac{\sqrt{2}\,\pi}{4}.$$

Part (ii)

Call the integral J. Its integrand is $\dfrac{1}{x} \cdot \dfrac{\sin x}{\sin x + \sin(1/x)}$. The denominator is safe: for $x \in [\frac{1}{2}, 2]$, both x and $1/x$ lie in $[\frac{1}{2}, 2] \subset (0, \pi)$, so $\sin x > 0$ and $\sin(1/x) > 0$.

Ask yourself Which substitution swaps $\sin x$ and $\sin\frac{1}{x}$ and keeps the limits $\frac{1}{2}$ and 2?

The answer is $x = 1/u$. Then $\mathrm{d}x = -\mathrm{d}u/u^2$. When $x = \frac{1}{2}$, $u = 2$, and when $x = 2$, $u = \frac{1}{2}$. The factor $\frac{1}{x}$ becomes u, so

$$J = \int_2^{1/2} u \cdot \frac{\sin(1/u)}{\sin(1/u) + \sin u} \cdot \Big(-\frac{\mathrm{d}u}{u^2}\Big) = \int_{1/2}^2 \frac{\sin(1/u)}{u(\sin u + \sin\frac{1}{u})}\,\mathrm{d}u.$$

Renaming u as x and adding this to the original form of J:

$$2J = \int_{1/2}^2 \frac{\sin x + \sin\frac{1}{x}}{x(\sin x + \sin\frac{1}{x})}\,\mathrm{d}x = \int_{1/2}^2 \frac{1}{x}\,\mathrm{d}x = \ln 2 - \ln\tfrac{1}{2} = 2\ln 2.$$

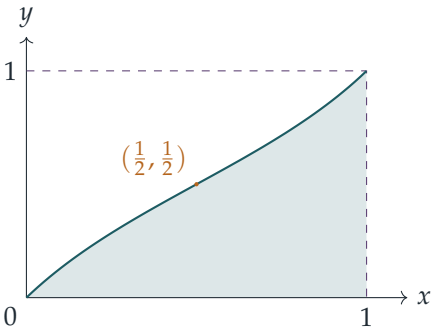
Hence

$$\int_{1/2}^2 \frac{\sin x}{x(\sin x + \sin\frac{1}{x})}\,\mathrm{d}x = \ln 2.$$

The factor $\frac{1}{x}$ is essential here. The measure $\frac{\mathrm{d}x}{x}$ is unchanged by $x \mapsto 1/x$ (apart from the orientation), in the same way that $\mathrm{d}x$ is unchanged by $x \mapsto a - x$.

Checks

- The integrand $g(x) = \frac{f(x)}{f(x)+f(a-x)}$ satisfies $g(x) + g(a-x) = 1$. Its graph is therefore symmetric under a half-turn about $(\frac{a}{2}, \frac{1}{2})$, so the area under it is half the area of the $a \times 1$ rectangle. The picture shows this for the first integral.



- Numerically, Simpson's rule gives 0.500000 for the first integral, 1.110721 for the second (and $\frac{\sqrt{2}\pi}{4} \approx 1.110721$), and 0.693147 for part (ii) (and $\ln 2 \approx 0.693147$).

Common slips

- Forgetting the factor $\sqrt{2}$ when rewriting $\sin(x + \frac{\pi}{4})$, which gives $\frac{\pi}{4}$ instead of $\frac{\sqrt{2}\pi}{4}$.
- In part (ii), losing track of the $\frac{1}{x}$: it turns into u, and combines with $-\frac{1}{u^2}$ to leave $-\frac{1}{u}$.

Beyond STEP

Both tricks exploit a symmetry of the interval that leaves the integral's "measure" unchanged: reflection $x \mapsto a - x$ preserves $\mathrm{d}x$, and inversion $x \mapsto 1/x$ preserves $\frac{\mathrm{d}x}{x}$. At university this becomes the idea of an invariant measure. For example, $\frac{\mathrm{d}x}{x}$ is the natural measure on the positive reals under multiplication, and it underlies the Mellin transform. For the real-analysis foundations of substitution, see Abbott, *Understanding Analysis*. For many more STEP-style integration tricks, see Siklos, *Advanced Problems in Mathematics*.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.